Analysis of viral genomes using the Galaxy platform

Exercise 2: Quality control and trimming

Despite the improvement of sequencing methods, there is no error-free technique. A correct measuring of the sequencing quality is essential to identify issues in the sequencing process; thus, this must be the first step in every sequencing analysis. Then, once the quality control is finished, it's important to remove low quality reads, or short reads, which is why a subsequent trimming step is mandatory. After the trimming step, it is recommended to perform a new quality control step to be sure that trimming worked correctly.

1. Illumina Quality control and trimming

Title	Pre-processing
Training dataset:	**PRJEB43037** - In August 2020, an outbreak of **West Nile Virus** affected 71 people with meningoencephalitis in Andalusia and 6 more people in Extremadura (south-west of Spain), causing a total of eight deaths. The virus belonged to the lineage 1 and was relatively similar to previous outbreaks occurred in the Mediterranean region. Here, we present a detailed analysis of the outbreak, including an extensive phylogenetic study. This is one of the outbreak samples.
Questions:	• How do I check whether my Illumina data was correctly sequenced? • How can I improve the quality of my data?
Objectives:	• Perform a quality control on raw Illumina reads. • Perform a quality and length-based trimming on raw Illumina reads. • Perform a quality control on trimmed Illumina reads.
Estimated time:	25 min

1.1. Quality control

1.1.1. Upload data

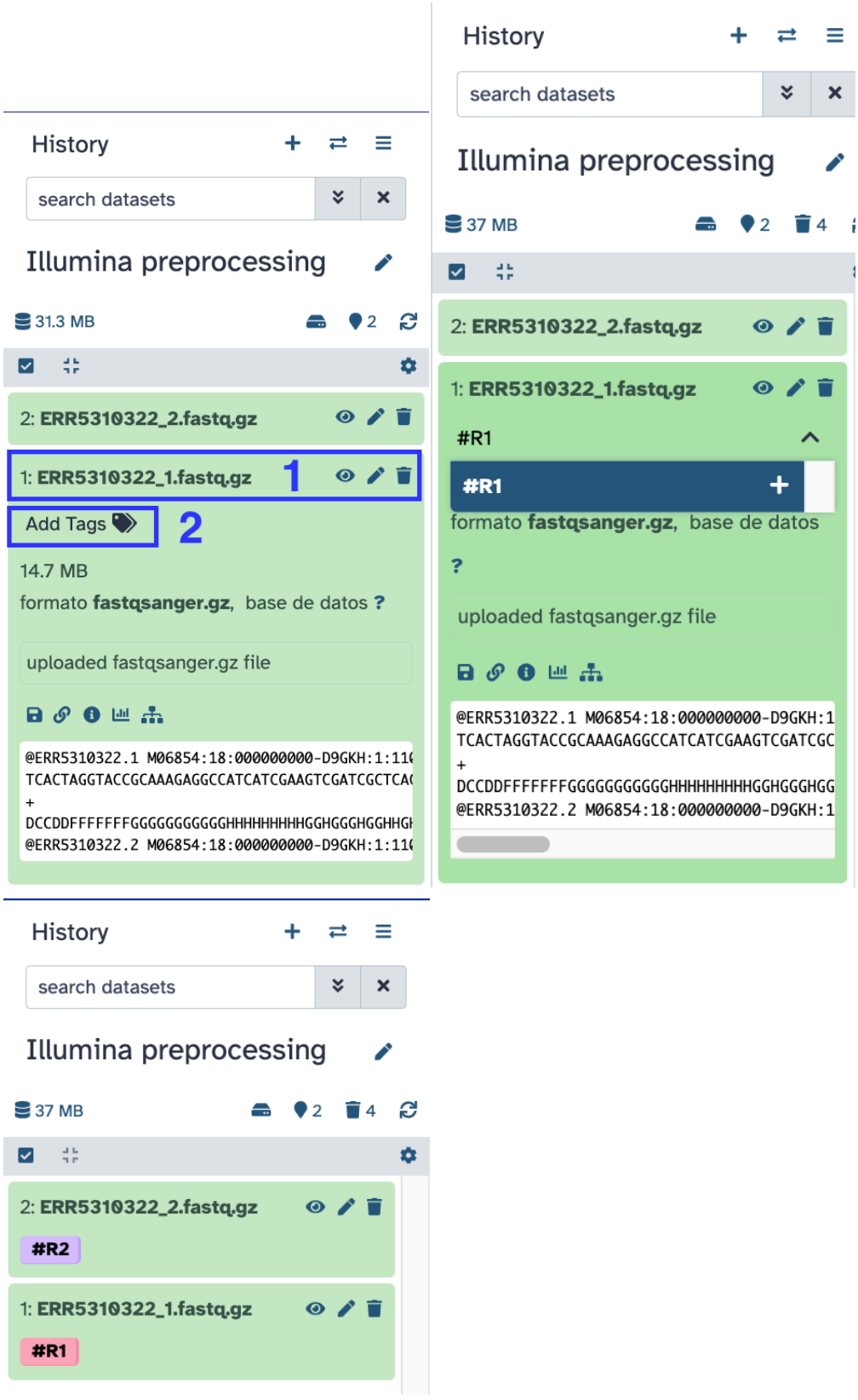
To run the quality control over the samples, follow these steps:

1. **Create a new history, as we explained yesterday** named **Illumina preprocessing**.
2. **Upload data as seen yesterday**, copy and paste the following URLs:

```
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_1.fastq.gz
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_2.fastq.gz
```

FASTQ files will be uploaded as **.fastqsanger.gz** files, which is a way Galaxy has to identify FASTQ files which follow the Phred+33 encoding.

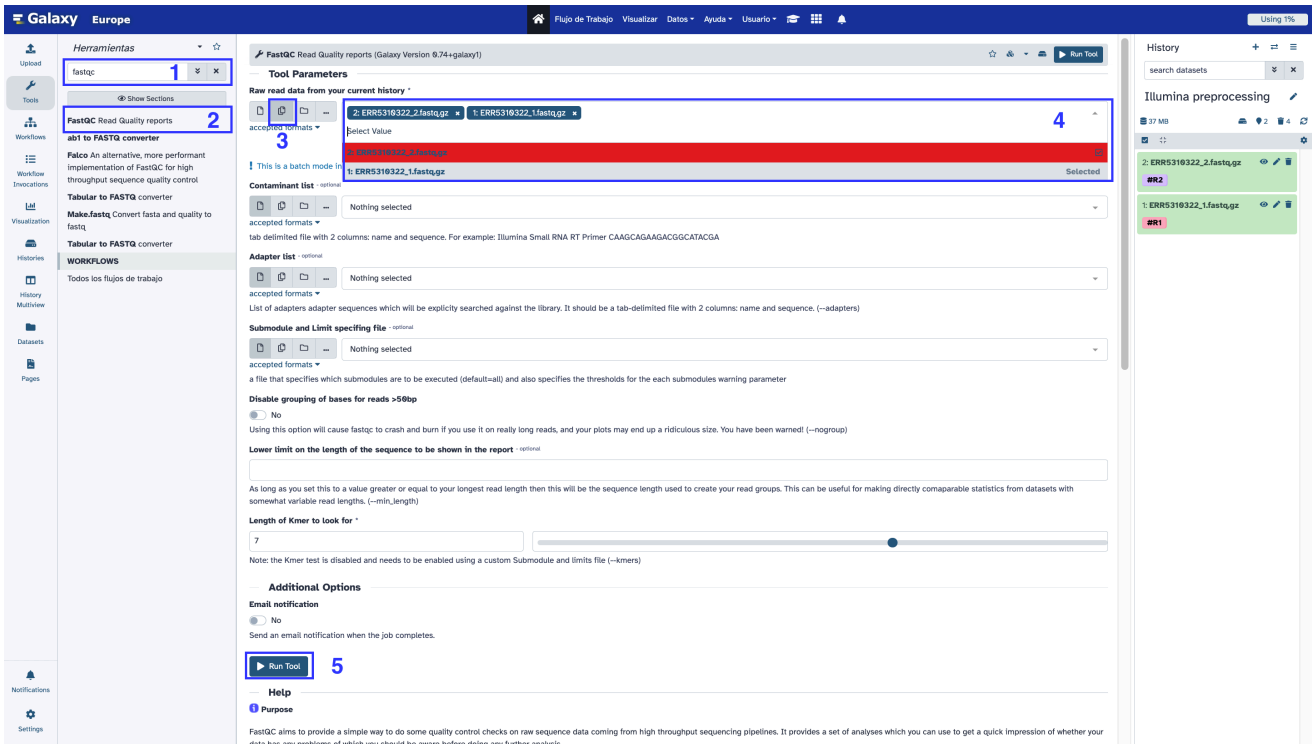
3. You can add some tags to the files. **It is mandatory that the tag starts with # to be propagated to the processes**.



1.1.2. Run FastQC

1. Search for the **FastQC** tool in the left panel.
2. Select **FastQC - Read Quality reports**.
3. Select multiple datasets in Raw read data from your current history.
4. Select the two datasets (R1 & R2).

5. Then go down and select **Run tool**.



FastQC will generate for each dataset two results: an HTML file with a variety of graphs (**WebPage**) and a TXT file with raw data used for each one of its modules (**RawData**). To see the results, we will open the jobs with **WebPage** in their name for both data 1 and data 2 (click on the **eye icon**).

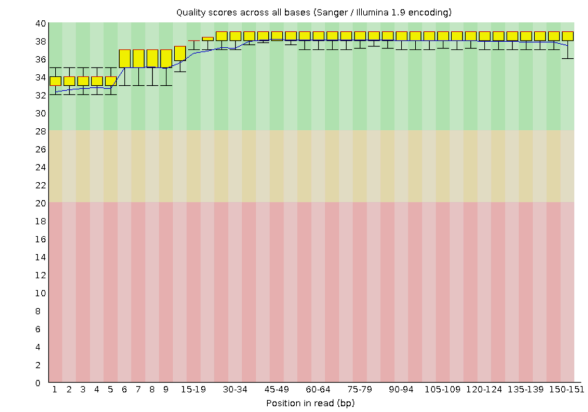
Basic Statistics

Measure	Value
Filename	ERR5310322_1_fastq_gz.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	265989
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	51

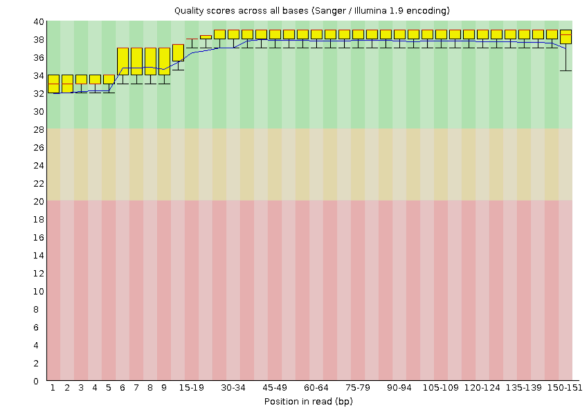
Basic Statistics

Measure	Value
Filename	ERR5310322_2_fastq_gz.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	265989
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	51

Per base sequence quality



Per base sequence quality



Here, you can see the number of reads in each file before trimming, the maximum and minimum length of all reads in the sample, and the quality plots for both R1 and R2. They look quite good, but we will next perform trimming over these samples.

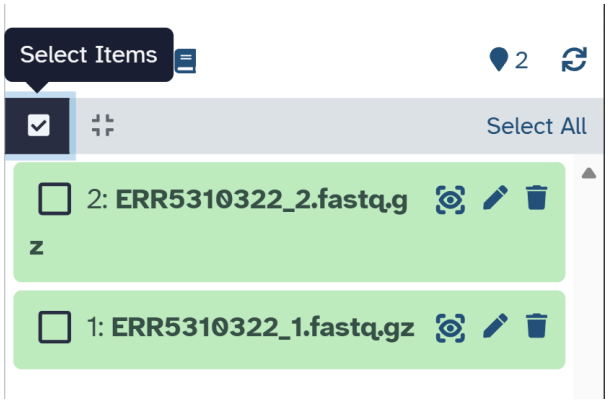
- How many reads do these samples have?
- Which tool can we use to check whether our Illumina data was correctly sequenced or not?

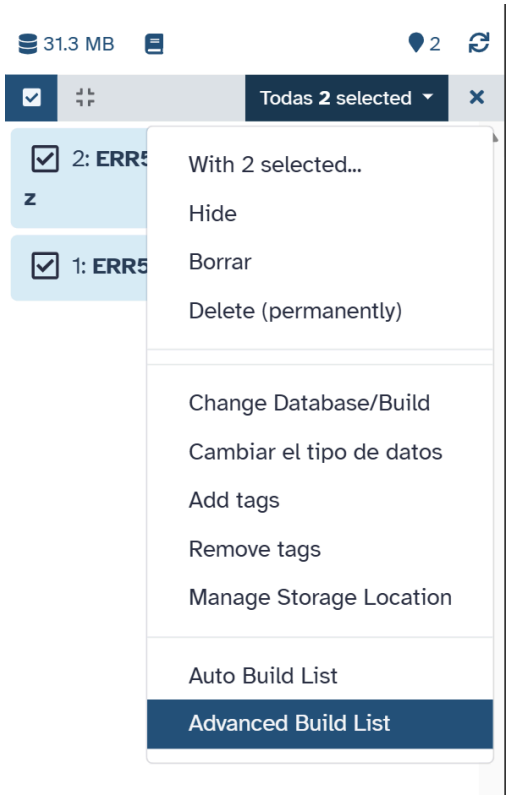
1.2. Trimming

Once we have performed the quality control, we can carry on with the **quality** and **read length trimming**:

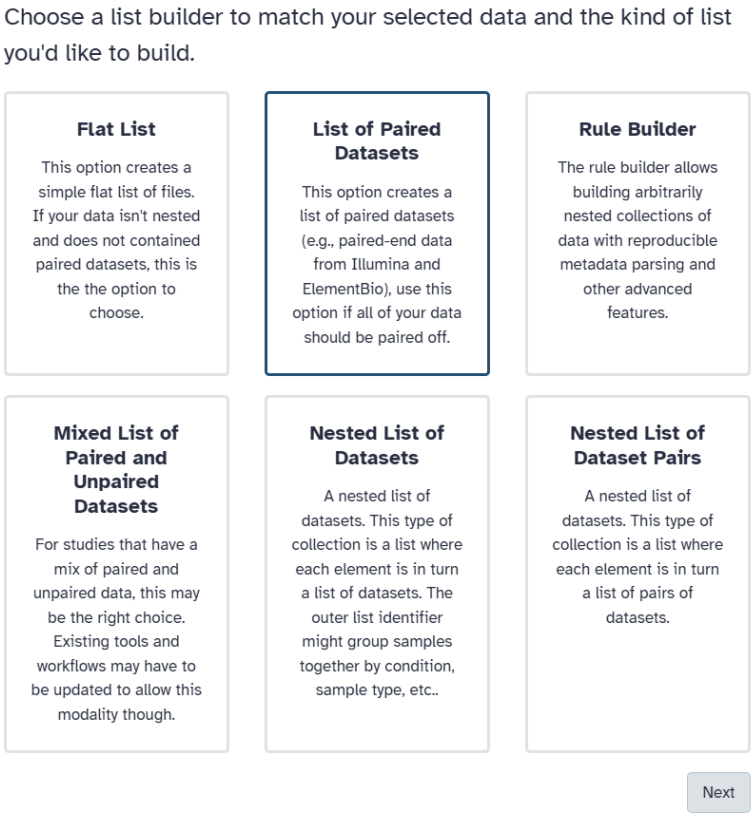
1.2.1. Run Fastp

1. Search for **fastp** in the tools.
2. Then select **fastp - fast all-in-one preprocessing for FASTQ files**.
3. Single-end or paired reads > **Paired Collection** (this is for most recent versions of the software; for older versions, you'd be able to select directly "Paired" and load each one of the datasets separately).

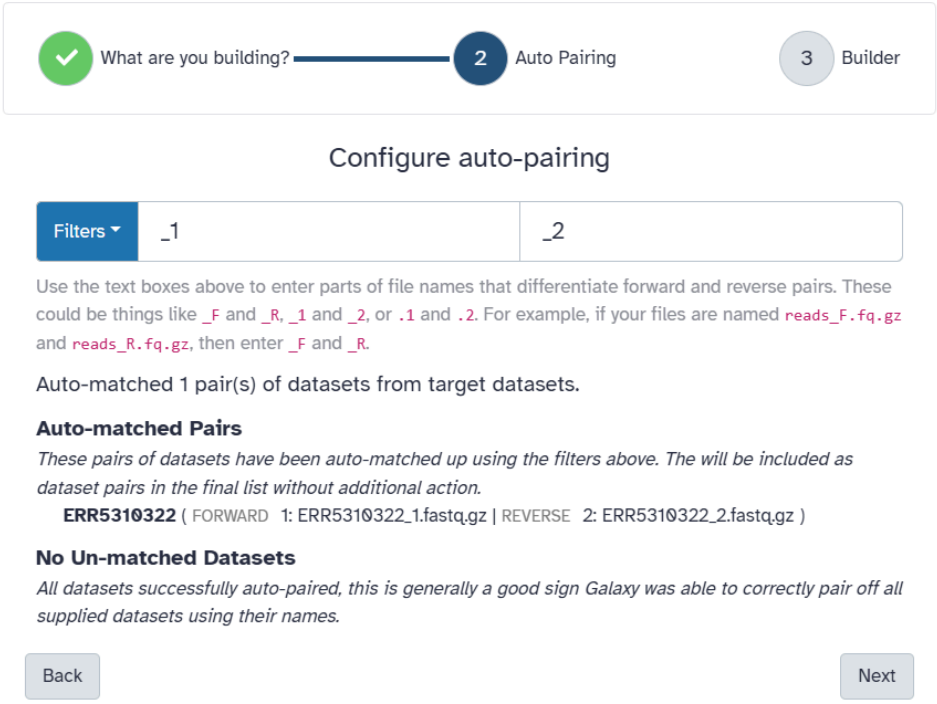
> ⚠ Warning
>
> To use this option from Fastp, it is necessary to first create a **list of paired datasets**, given our .fastq files, which will then be used for the rest of the steps of this workshop. In order to do so, we must:
>
> 1. **Select our fastqsanger.gz files that we uploaded in the beginning, and then click on 2 of X selected**.
>
> 
>
>
> 2. **Select Advanced Build List**.



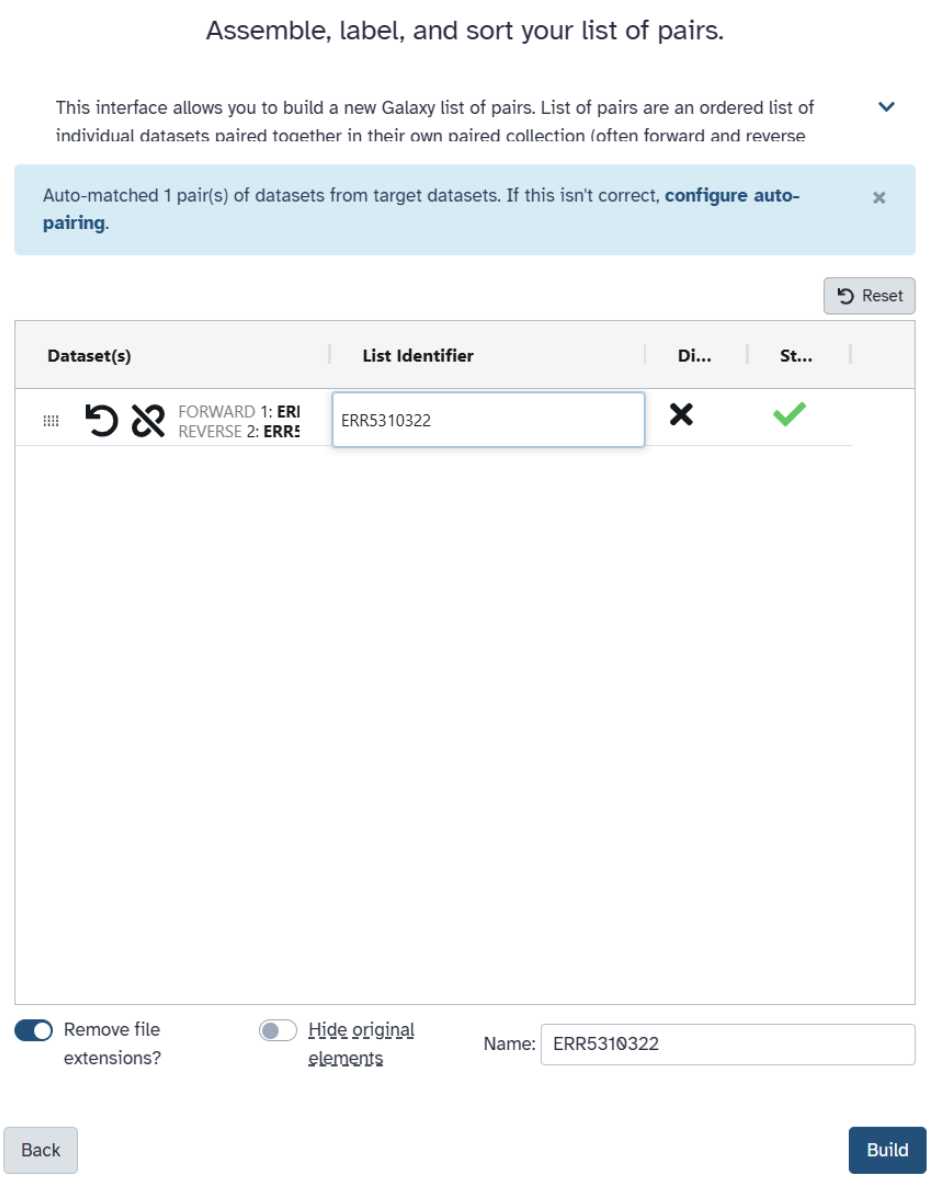
3. **Select the List of Paired Datasets box and click on *Next*.**



4. **Check the auto-pairing configuration and click again on *Next*.**



5. **Give a name to your list and finally click on *Build***.



After having done this, a list of datasets will have been generated:

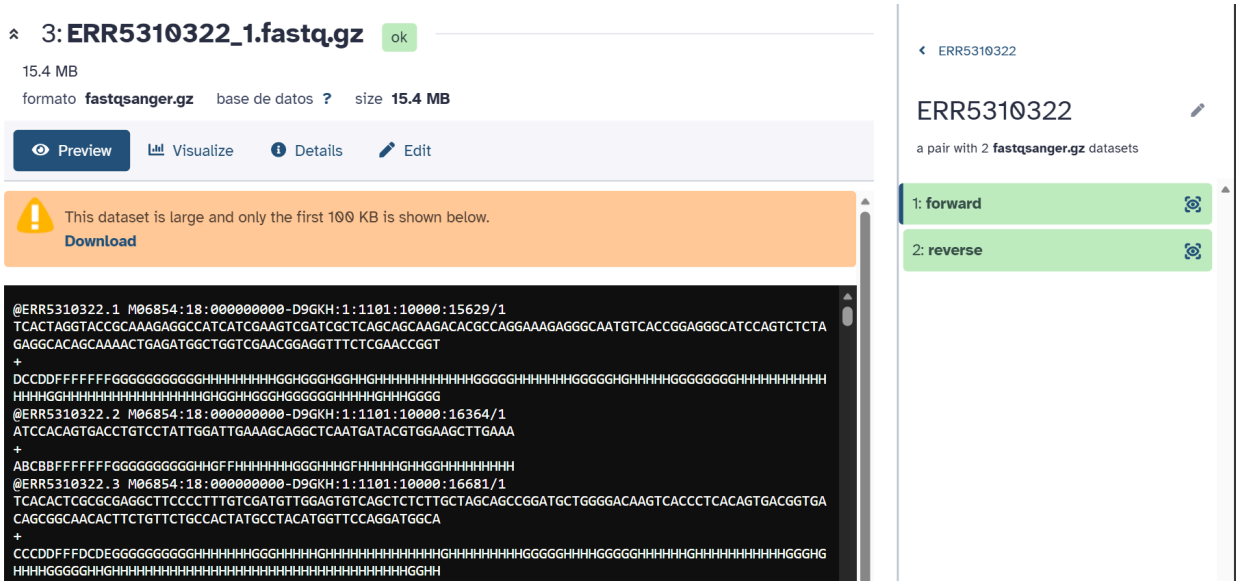
5: **ERR5310322**

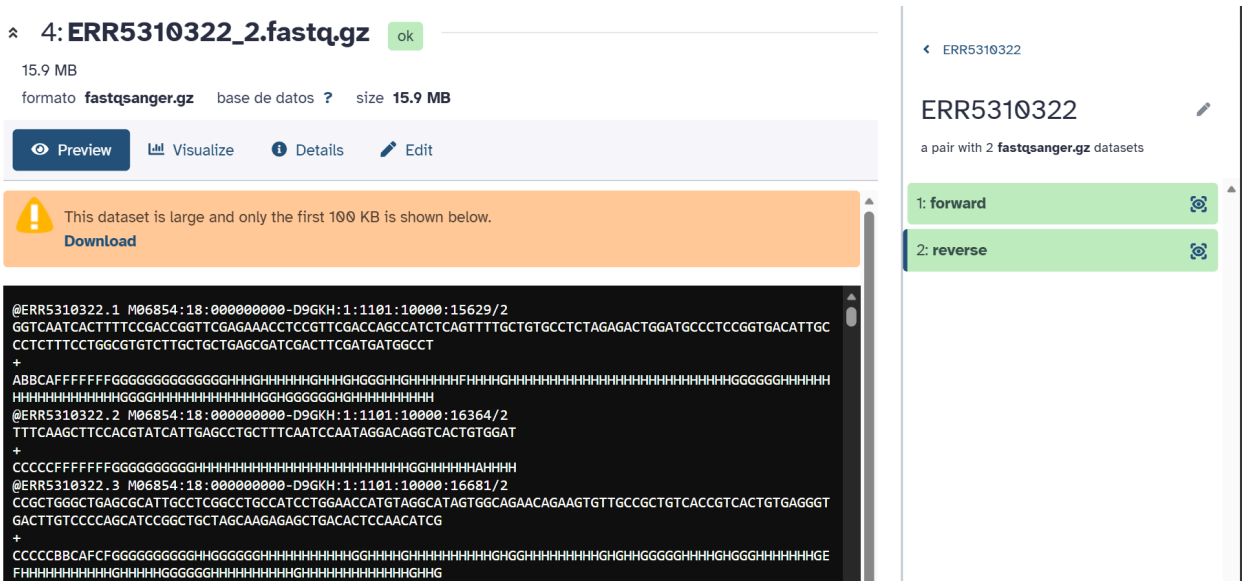
a list with 1 **fastqsanger.gz** pair

1: **ERR5310322**

a pair with 2 **fastqsanger.gz** datasets

This list will have the forward and reverse reads:





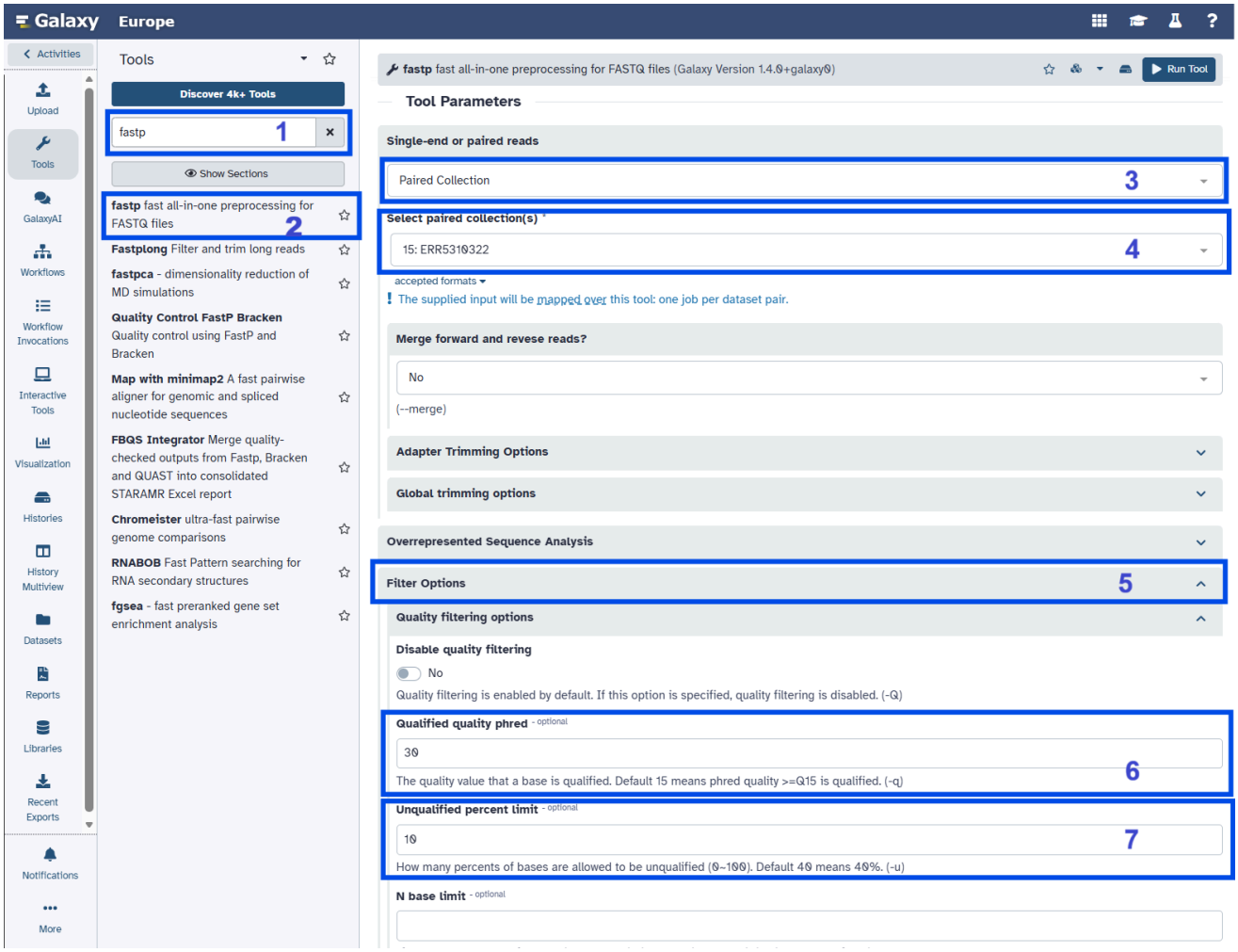
Once this is done, you'll be able to select your list as input of Fastp.

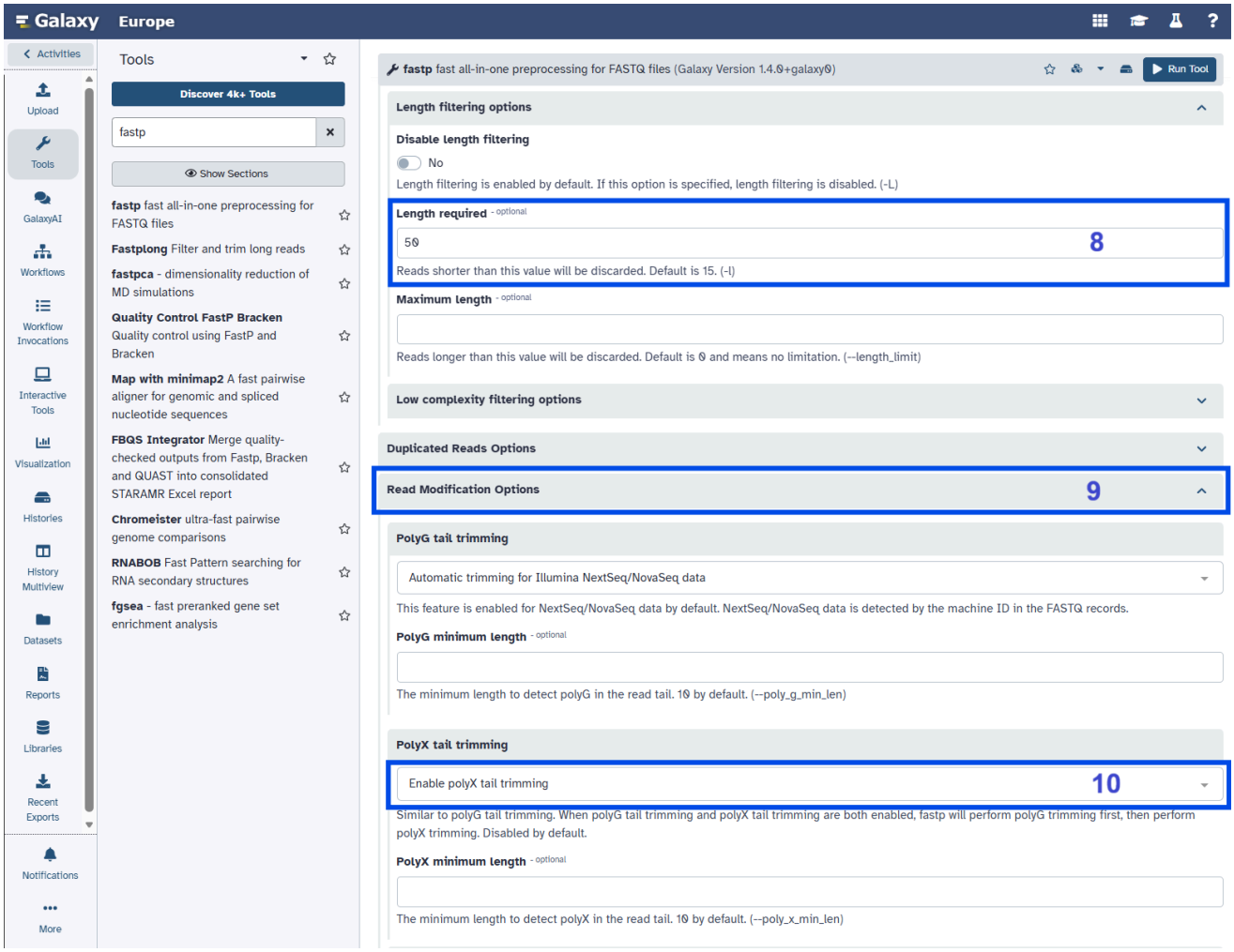
1. Display **Filter Options** (for this exercise, we will not provide adapter sequences and we will not trim X bases in front or tail, so ignore the Adapter Trimming Options and Global Trimming Options).
    - **Quality Filtering options**
        - Qualified Quality Phred = **30**
        - Unqualified percent limit = **10**
        - N base limit = - (will be used with its default value (5))
    - Length Filtering Options
        - Length required = **50**
2. **Read modification options**
    - PoliX tail trimming > **Enable polyX tail trimming**
    - Per read cutting by quality options

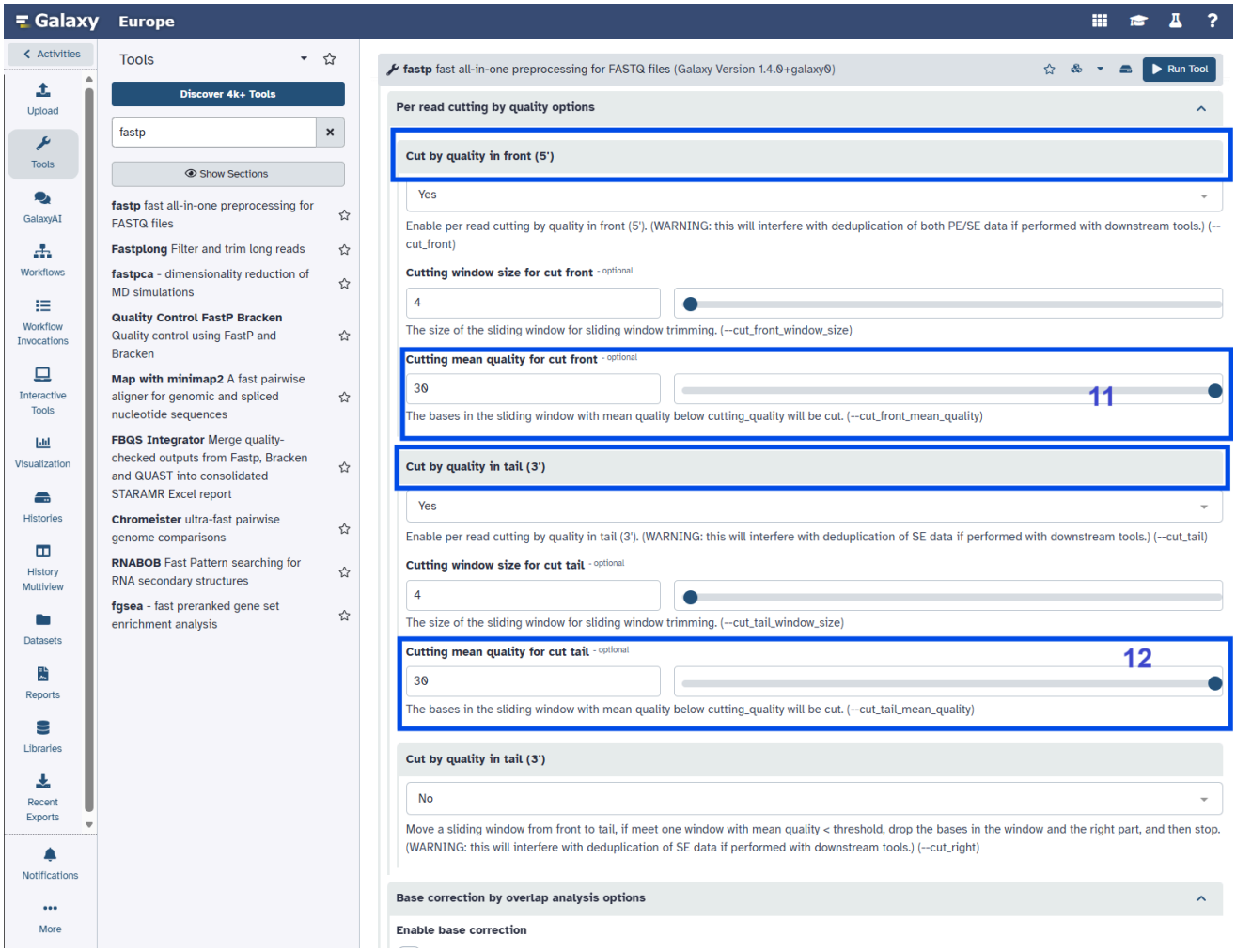
- Cut by quality in front (5') > **Yes**
- Cut by quality in tail (3') > **Yes**
- Cutting mean quality (for both) = **30**

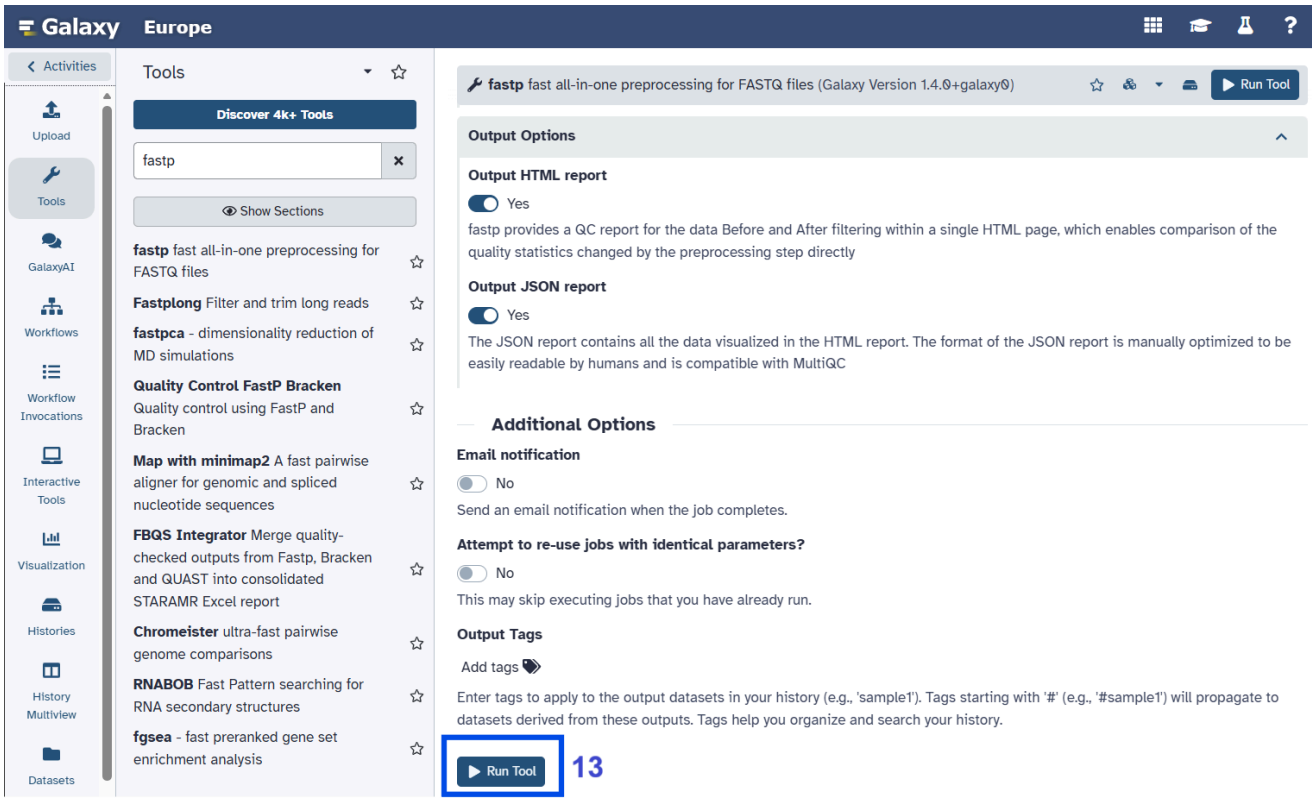
> ⚠ Warning
>
> There are two sections called "Cut by quality in tail (3')" in Galaxy for fastp. **The first one is the one that should be enabled for this exercise**, since this process takes a sliding window of N bases (4 by default) from the 3' end and removes windows whose mean quality is below the quality threshold, until a window with good mean quality is found.
>
> The other process, on the other hand, is more aggresive, since it takes a sliding window from the 5' end and, if it meets one window with mean quality < threshold, it drops the bases in such window and the whole right part, and then stops.

3. Finally, click on **Run tool**.




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fastpca - dimensionality reduction of MD simulations
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Map with minimap2 A fast pairwise aligner for genomic and spliced nucleotide sequences
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Chromeister ultra-fast pairwise genome comparisons
RNABOB Fast Pattern searching for RNA secondary structures
fgsea - fast preranked gene set enrichment analysis
fastp fast all-in-one preprocessing for FASTQ files (Galaxy Version 1.4.0+galaxy0)
Run Tool
Length filtering options
Disable length filtering
No
Length filtering is enabled by default. If this option is specified, length filtering is disabled. (-L)
Length required - optional
50
8
Reads shorter than this value will be discarded. Default is 15. (-l)
Maximum length - optional
Reads longer than this value will be discarded. Default is 0 and means no limitation. (--length_limit)
Low complexity filtering options
Duplicated Reads Options
Read Modification Options
9
PolyG tail trimming
Automatic trimming for Illumina NextSeq/NovaSeq data
This feature is enabled for NextSeq/NovaSeq data by default. NextSeq/NovaSeq data is detected by the machine ID in the FASTQ records.
PolyG minimum length - optional
The minimum length to detect polyG in the read tail. 10 by default. (--poly_g_min_len)
PolyX tail trimming
Enable polyX tail trimming
10
Similar to polyG tail trimming. When polyG tail trimming and polyX tail trimming are both enabled, fastp will perform polyG trimming first, then perform polyX trimming. Disabled by default.
PolyX minimum length - optional
The minimum length to detect polyX in the read tail. 10 by default. (--poly_x_min_len)


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fastp fast all-in-one preprocessing for FASTQ files (Galaxy Version 1.4.0+galaxy0)
Run Tool
Per read cutting by quality options
Cut by quality in front (5')
Yes
Enable per read cutting by quality in front (5'). (WARNING: this will interfere with deduplication of both PE/SE data if performed with downstream tools.) (--cut_front)
Cutting window size for cut front - optional
4
The size of the sliding window for sliding window trimming. (--cut_front_window_size)
Cutting mean quality for cut front - optional
30
11
The bases in the sliding window with mean quality below cutting_quality will be cut. (--cut_front_mean_quality)
Cut by quality in tail (3')
Yes
Enable per read cutting by quality in tail (3'). (WARNING: this will interfere with deduplication of SE data if performed with downstream tools.) (--cut_tail)
Cutting window size for cut tail - optional
4
The size of the sliding window for sliding window trimming. (--cut_tail_window_size)
Cutting mean quality for cut tail - optional
30
12
The bases in the sliding window with mean quality below cutting_quality will be cut. (--cut_tail_mean_quality)
Cut by quality in tail (3')
No
Move a sliding window from front to tail, if meet one window with mean quality < threshold, drop the bases in the window and the right part, and then stop. (WARNING: this will interfere with deduplication of SE data if performed with downstream tools.) (--cut_right)
Base correction by overlap analysis options
Enable base correction


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fgsea - fast preranked gene set enrichment analysis
fastp fast all-in-one preprocessing for FASTQ files (Galaxy Version 1.4.0+galaxy0)
Run Tool
Output Options
Output HTML report
Yes
fastp provides a QC report for the data Before and After filtering within a single HTML page, which enables comparison of the quality statistics changed by the preprocessing step directly
Output JSON report
Yes
The JSON report contains all the data visualized in the HTML report. The format of the JSON report is manually optimized to be easily readable by humans and is compatible with MultiQC
Additional Options
Email notification
No
Send an email notification when the job completes.
Attempt to re-use jobs with identical parameters?
No
This may skip executing jobs that you have already run.
Output Tags
Add tags
Enter tags to apply to the output datasets in your history (e.g., 'sample1'). Tags starting with '#' (e.g., '#sample1') will propagate to datasets derived from these outputs. Tags help you organize and search your history.
Run Tool
13

Fastp will generate the trimmed reads along with a report in both JSON and HTML formats. To see the trimming stats, let's take a look at the **fastp on collection X: HTML report** file. You should see something like this:

ERR5310322.fastqsanger.gz

Summary

General

fastp version:	1.4.0 (https://github.com/OpenGene/fastp)
sequencing:	paired end (151 cycles + 151 cycles)
mean length before filtering:	105bp, 105bp
mean length after filtering:	113bp, 113bp
Insert size peak:	84

Before filtering

total reads:	531.978000 K
total bases:	56.257825 M
Q20 bases:	54.842431 M (97.484094%)
Q30 bases:	54.605191 M (97.062393%)
Q40 bases:	50.780000 K (0.090263%)
GC content:	50.644494%

After filtering

total reads:	433.314000 K
total bases:	49.003611 M
Q20 bases:	48.876432 M (99.740470%)
Q30 bases:	48.825481 M (99.636496%)
Q40 bases:	50.558000 K (0.103172%)
GC content:	51.087943%

▶ How many reads have we lost?

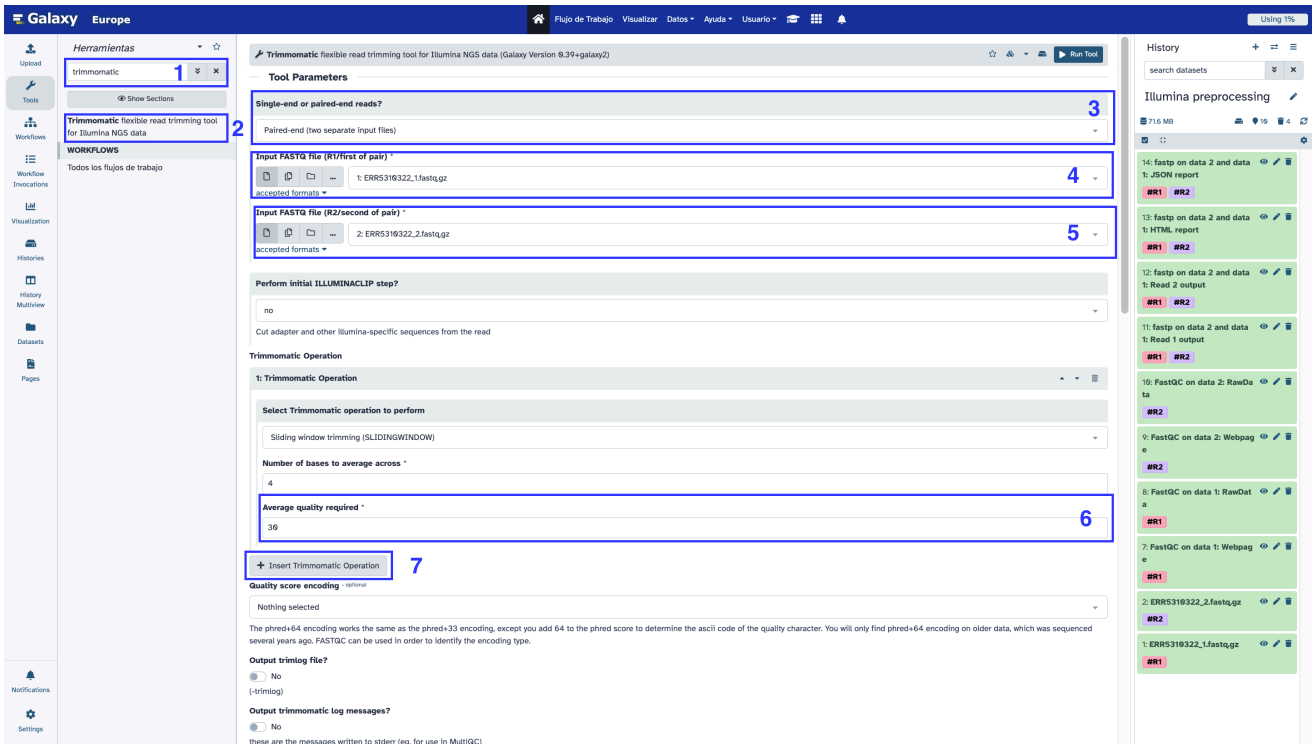
1.2.2. Other trimming tools: Trimmomatic

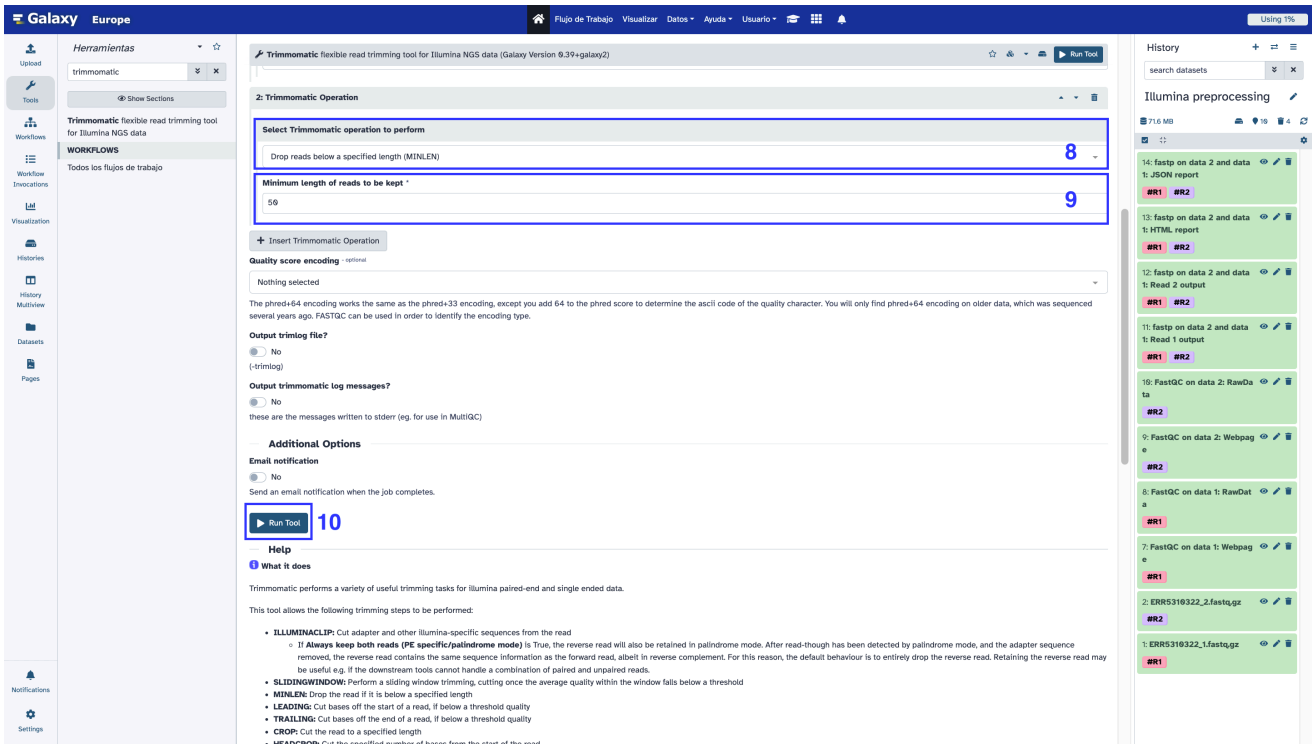
1. Search for **Trimmomatic** in the tools.
2. Select **Trimmomatic - flexible read trimming tool for Illumina NGS data**.
3. Single-end or paired-end reads? > **Paired-end (two separate input files)**.
4. Average quality required = **30**.
5. Insert Trimmomatic Operation:

- Select Trimmomatic operation to perform: **MINLEN**.
- Minimum length of reads to be kept = **50**.

6. Select **Run tool**. Trimmomatic will generate paired and unpaired .fastqsanger.gz files for both forward and reverse reads.

- Two .fastqsanger.gz files (R1-paired and R2-paired) contain reads that survived filtering.
- Additionally two .fastqsanger.gz files (R1-unpaired and R2-unpaired) contain reads where one of the pair failed the filtering steps.





Trimmomatic **does not** perform statistics over trimmed reads, so we need to perform FastQC again over the Trimmomatic results.

▶ Try to do it on your own.

▶ How can we improve the quality of our data?

> **This hands-on history URL: https://usegalaxy.eu/u/svarona/h/illumina-preprocessing**

2. Nanopore Quality control and trimming

Title	Galaxy
Training dataset:	The data we will manage now corresponds to Nanopore amplicon sequencing data using ARTIC network primers for the SARS-CoV-2 genome. From the Fast5 files generated by the ONT software, we will select the **PASS** reads, so they are already filtered by quality.
Questions:	• How do I know if my Nanopore data was correctly sequenced?
Objectives:	• Perform a quality control on raw Nanopore reads. • Perform a length-based trimming on raw Nanopore reads. • Perform a quality control on trimmed Nanopore reads.
Estimated time:	15 min

2.1. Quality control

To run the quality control over the samples, follow these steps:

1. **Create a new history has explained yesterday** named **Nanopore quality**.
2. **Upload data as seen yesterday**, copy and paste the following URLs:

```
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minior
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minior
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minior
```

2.1.1. PycoQC

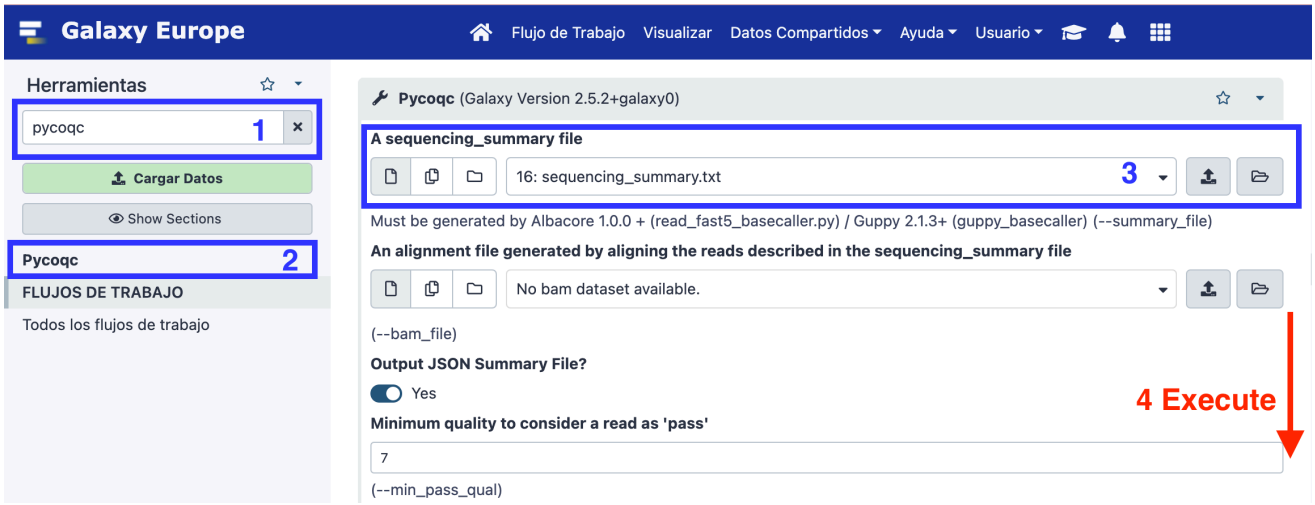
To use PycoQC we need to use the `sequencing_summary.txt` provided by the Nanopore sequencing machine (this file does not contain read sequences, but other relevant information, such as read length, read ID, Phred quality, etc., in this case associated with 9 barcodes (the one sample that we uploaded previously along with other 8)).

Upload data as seen yesterday, copy and paste the following URL:

```
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minior
```

1. Search for the **Pycoqc** tool.

2. Select **Pycoqc - quality control for Nanopore sequencing data**.
3. In *A sequencing_summary file*: Select the `sequencing_summary.txt` we just uploaded.
4. Select **Run tool**.



Then we can inspect the resulting **PycoQC HTML Report**:

General run summary

Status	Run Duration (h)	Active Channels	Number of Runids	Number of Barcodes
All Reads	7.98	496	1	9
Pass Reads	7.98	494	1	9

Basecall summary

Status	Reads	Bases	N50	Median Read Length	Median PHRED score
All Reads	2.449000e+4	1.284332e+7	515	514	13.055
Pass Reads	2.435200e+4	1.276470e+7	515	514	13.067

Questions

- How many reads does each sample have?
- Do you understand all the plots?
- How can we check whether my Nanopore data was correctly sequenced?

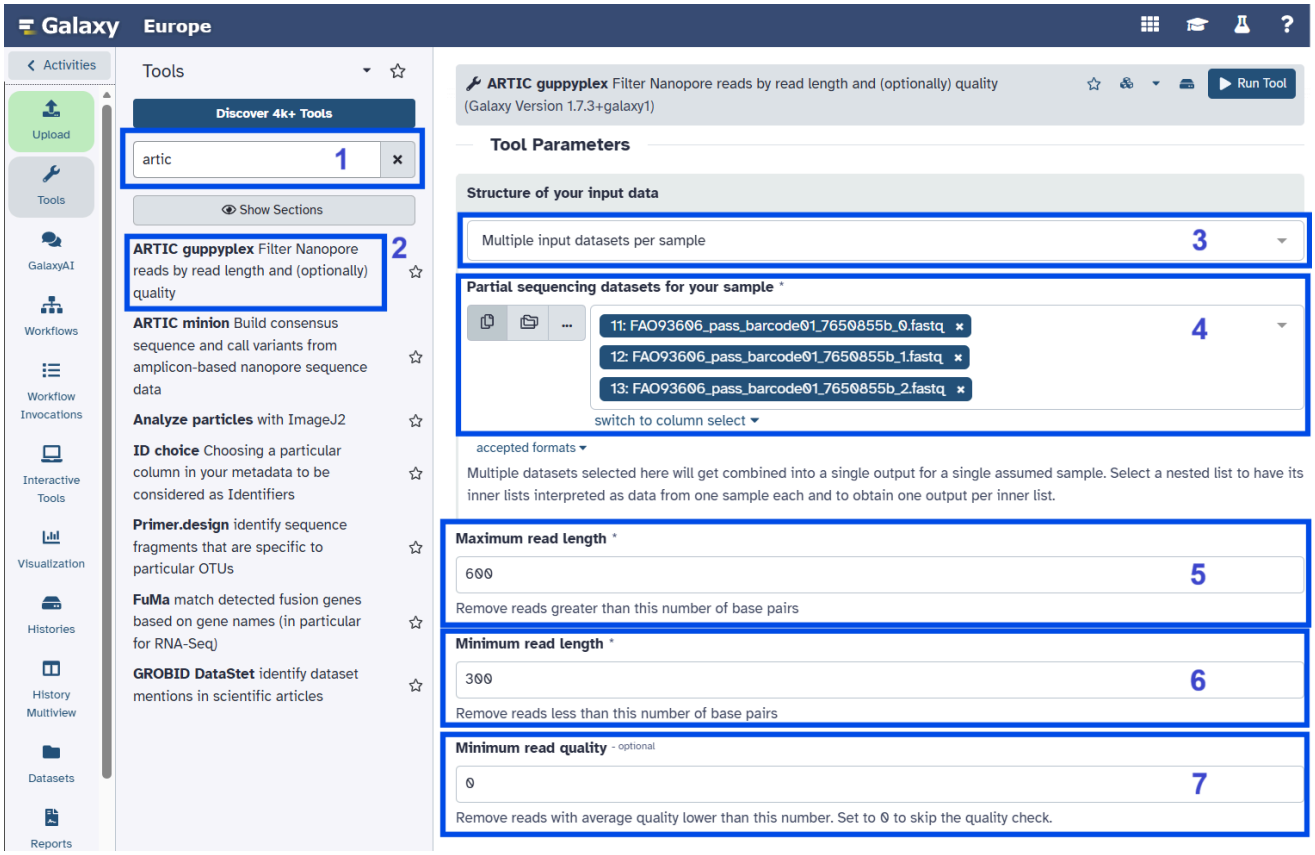
2.2. Trimming

When Nanopore reads are being sequenced, the **MinKNOW** software splits Fast5 reads into quality **PASS** and quality **FAIL**. As we will select only Fast5 **PASS reads**, we won't need to perform a quality trimming, so even if we see that the reads have a bad Phred score, we know that the ONT software considered the reads as "**good quality**".

Therefore, we will only be performing **read length trimming**. Since we are using amplicon sequencing data, we won't be expecting reads neither smaller than 400 nucleotides, nor higher than 600, which would obviously correspond to **chimeric reads** (sequences from two or more distinct DNA fragments joined together).

2.2.1. ARTIC

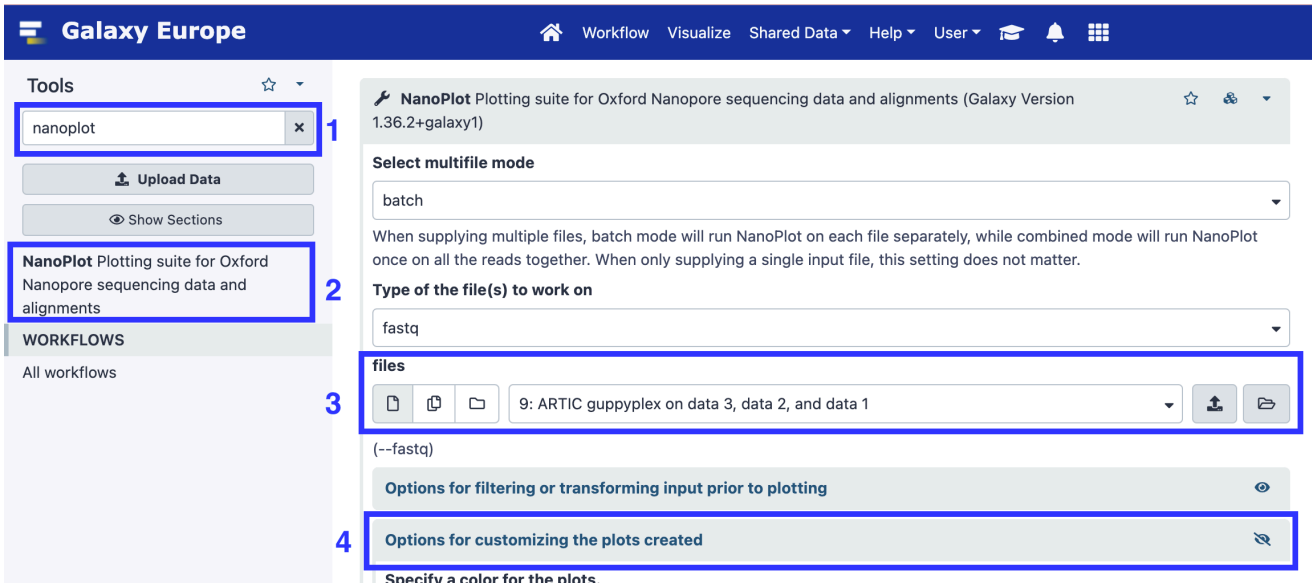
1. Search for **artic** tool.
2. Select **ARTIC guppyplex - Filter Nanopore reads by read length and (optionally) quality**.
3. Structure of your input data: **Multiple input datasets per sample**.
4. While pressing the *Ctrl* key, **select the three samples**.
5. Remove reads longer than = **600**.
6. Remove reads shorter than = **300**.
7. Minimum read quality = **0** (**we had already selected pass reads**).
8. **Run Tool**. ARTIC will generate the trimmed reads in a separate output file.

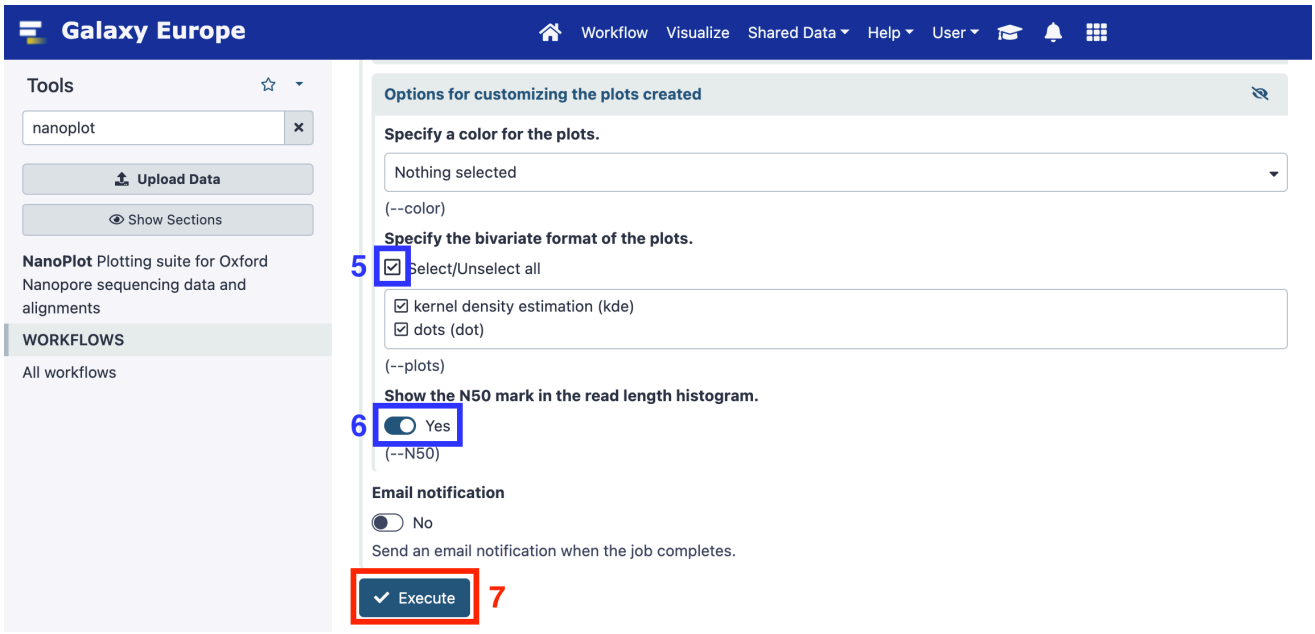


2.2.2. NanoPlot

Now we will run NanoPlot on the reads filtered by length after running ARTIC, with the objective of plotting post-processed reads (NanoPlot can also apply filters based on quality and length, but this time we will use it only for plotting purposes, since we have already filtered based on these criteria):

1. Search for the **Nanoplot** tool and select **NanoPlot - Plotting suite for Oxford Nanopore sequencing data and alignments**
2. Run the tool as follows:
    - In the *files* part, select the **ARTIC output file** (ARTIC guppyplex on data set XX - XX).
    - Display **Options for customizing the plots created**:
        - **Specify the bivariate format of the plots** > Select all
        - **Show the N50 mark in the read length histogram** > Yes
3. Select **Run Tool**. An HTML report will be generated along with the corresponding statistics.





Questions

- Did our data length and quality improve?
- How many reads did we lose during the trimming step?

> **This hands-on history URL**: https://usegalaxy.eu/u/svarona/h/nanopore-quality.